Written exam 15 December, 2023

Course title: Network Optimization **Course no:** 42115

Aids allowed: All

Exam duration: 4 hours

Weighting: The points are indicated at each question

Language: You can write in Danish or English

Figures: Figures can be found on the last pages of this assignment. You can use them in your answer. Pull out the individual pages, and insert them in your answer.

Critical path with earliest start times

Thristur (þristur) is the favorite mini-candy in Iceland. Production consists of the following six steps:

activity	description	duration	predecessor	earliest starting time
A	Cut toffee into squares	2	F	8
B	Cut licorice into pieces	1		
C	Cover with chocolate	3	A, E	
D	Mix caramel cream	3		
E	Melt chocolate	4	B, F	7
F	Mix caramel cream and licorice	4	D, B	

Since some of the ingredients are made in remote factories, we have earliest starting times on some of the activities.

Problem 1 (2 point) Formulate the earliest starting times as tasks. Show the resulting table of tasks. ■

Problem 2 (1 point) Sort the tasks in topological order (either as graph, or as a list). ■

Problem 3 (3 point) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan T. Show the recursion you are using. (**Note:** if you did not solve problem 1, then solve the original critical path problem). ■

Problem 4 (1 point) Indicate the critical path, and argue why the edges are on the critical path. ■

Transportation problem

Two bakeries B_1 and B_2 are delivering cakes to three courses C_1 (42115), C_2 (42114) and C_3 (42142). The bakeries are producing 6 and 4 cakes respectively. Course C_1 can eat at most 5 cakes, course C_2 can eat at most 3 cakes, and course C_3 can eat at most 4 cakes. All cakes produced need to be eaten. The transportation costs are given by the following table

	C_1	C_2	C_3	production
B_1	2	3	-	6
B_2	1	-	7	4
max demand	5	3	4	

A "-" indicates that transportation is not possible. The objective is to minimize overall transportation costs. **Note**, that the problem is not balanced.

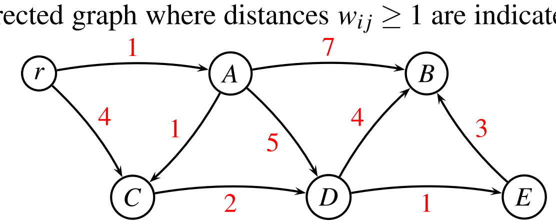
Problem 5 (1 point) Formulate the problem as a transportation problem, by specifying the network and all costs / demands. ■

Problem 6 (2 point) Find a greedy solution for the transformed problem: Consider the bakeries in the order $B_1, B_2, \ldots$ always choosing the cheapest option for transporting the cakes. ■

Problem 7 (4 point) Find an optimal solution using the network simplex algorithm for the transportation problem. In each iteration, give sufficient details to follow the algorithm. Describe the termination criteria, and report the optimal solution. ■

Shortest Path Problem

Consider the following directed graph where distances $w_{ij} \geq 1$ are indicated with red.



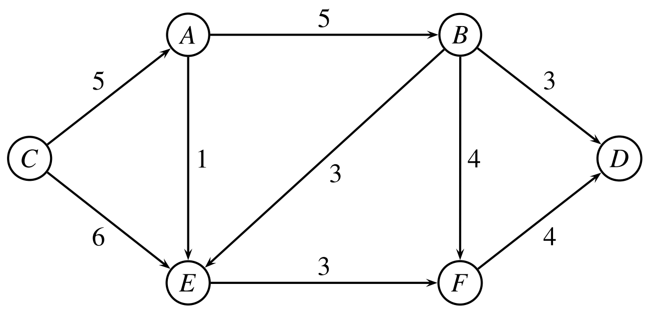
We want to find the shortest path from r to all nodes. If several paths have the same length, we are interested in the shortest path containing *most* edges.

Problem 8 (2 point) Formulate the problem as an ordinary shortest path problem. Show the resulting graph and edge costs. ■

Problem 9 (3 point) Solve the shortest path problem using Dijkstra's algorithm. Give sufficient detail to follow the calculations. Report the shortest path tree. (**Note:** if you could not solve the previous question, then solve the original shortest path problem). ■

Circulation problem / Maximum flow with demands

Given a directed graph $G = (V, E)$ and associated edge capacities u_{ij} for each edge (i, j). In the *Circulation Problem* we have multiple sources and multiple sinks. If a node has positive demand $d_i > 0$ it is called a sink. If a node has negative demand $d_i < 0$ it is called a source. Obviously, we need to have $\sum_{i \in V} d_i = 0$ to match supply and demand.



Assume that we have demands

node i	A	B	C	D	E	F
demand d_i	-2	0	-4	5	0	1

Problem 10 (2 point) Formulate the above problem as an ordinary maximum flow problem. (**Hint:** take inspiration from the min-cost flow problem.) ■

Assume that we have a lower bound on edge (B, E) of value $\ell_{BE} = 2$. This means that in a given solution, there should be at least 2 units flowing on the edge.

Problem 11 (2 point) Formulate the problem with lower bound on edge (B, E) as an ordinary maximum flow problem. ■

Problem 12 (4 point) Solve the resulting maximum flow problem. Provide sufficient details to follow your calculations. When choosing an augmenting path, always choose the path with largest possible capacity. (**Note:** if you could not solve the previous question, then solve the original problem.) ■

Problem 13 (1 point) Report the value of the maximum flow, and show a minimum cut. ■

Hungarian Algorithm

In the *Assignment problem* we are asked to assign a number of resources to an equal number of activities, so as to minimize total cost of allocation. Each resource should be assigned to exactly one activity, and each activity should have exactly one resource assigned.

The assignment problem is frequently solved with the *Hungarian Algorithm:* Given a cost matrix $C = \{c_{ij}\}$ saying how much it costs to assign resource i to activity j. The first step of the algorithm is to subtract the smallest number from each column, then subtract the smallest number from each row. (See example below).

$$\begin{pmatrix} - & 3 & 93 & 15 & 46 & 13 \\ 6 & - & 79 & 46 & 36 & 22 \\ 32 & 4 & - & 25 & 16 & 19 \\ 53 & 104 & 92 & - & 83 & 25 \\ 6 & 24 & 66 & 13 & - & 7 \\ 3 & 88 & 18 & 48 & 105 & - \end{pmatrix} \rightarrow \begin{pmatrix} - & 0 & 75 & 2 & 30 & 6 \\ 0 & - & 58 & 30 & 17 & 12 \\ 29 & 1 & - & 12 & 0 & 12 \\ 32 & 83 & 58 & - & 49 & 0 \\ 3 & 21 & 48 & 0 & - & 0 \\ 0 & 85 & 0 & 35 & 89 & - \end{pmatrix}$$

Next, we need to find the largest set of *independent* 0-values. Two 0-values are *independent* if they do not appear in the same row or column. For instances, in the first matrix below the bold-face 0-values are independent, but not in the second matrix

$$\begin{pmatrix} . & \mathbf{0} & . & . & . & . \\ \mathbf{0} & . & . & . & . & . \\ . & . & . & . & \mathbf{0} & . \\ . & . & . & . & . & \mathbf{0} \\ . & . & . & 0 & . & 0 \\ 0 & . & \mathbf{0} & . & . & . \end{pmatrix} \qquad \begin{pmatrix} . & \mathbf{0} & . & . & . & . \\ 0 & . & . & . & . & . \\ . & . & . & . & 0 & . \\ . & . & . & . & . & \mathbf{0} \\ . & . & . & 0 & . & \mathbf{0} \\ \mathbf{0} & . & \mathbf{0} & . & . & . \end{pmatrix} \tag{1}$$

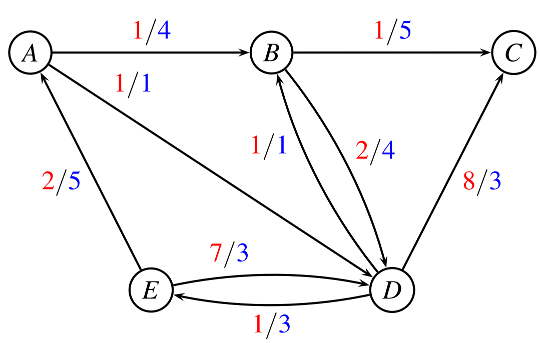
Problem 14 (4 point) Given a matrix with 0-values as in (1). Formulate the problem of finding the largest set of independent 0-values as a maximum flow problem. Explain how the graph is constructed, and how a solution should be interpreted. ■

Repositioning of empty containers

The below instance, MEDITLINE, shows a small liner shipping network. Nodes represent ports, and edges represent legs sailed by vessels. The capacities indicate the remaining capacity of the vessels after some mandatory orders have been fulfilled. Therefore, the capacity may vary for each leg.

In mathematical terms, we are given a weighted oriented graph $G = (V, E, c, u)$ where each edge $(i, j) \in E$ has an associated cost (pr. container) $c_{ij} \geq 0$, and a capacity $u_{ij} \geq 0$.

MEDITLINE **(Route net of a liner shipping company)**



For edge (i, j) the first number (red) is the cost c_{ij} and the second number (blue) is the capacity u_{ij}.

Consider the MEDITLINE network from the project assignment. In a given month, we have the following inventory of empty containers:

node	A	B	C	D	E
inventory	3	4	2	4	3

All empty containers are equal. We want to reposition the empty containers such that all containers are moved to C and D. Furthermore, the two nodes C and D should have the same amount of containers after repositioning.

Problem 15 (2 point) Formulate the repositioning of empty containers in MEDITLINE as a *minimum cost flow problem*. Define the graph and the demands in each node. ■

Problem 16 (5 point) Solve the MEDITLINE repositioning problem, using a network algorithm. State which algorithm you use, and give sufficient details to make it possible to follow your calculations. Report the optimal solution. ■

Flowing containers

We consider again the route net MEDITLINE from project assignment 2023, in which we have *new demands* as follows:

k	s_k	t_k	d_k
1	E	D	3
2	D	C	1
3	E	C	2

Commodity $k=1$ is shipping three containers from E to D, commodity $k=2$ is shipping one container from D to C, and commodity $k=3$ is shipping two containers from E to C.

We are considering a *unsplittable multi-commodity flow problem* in which all the d_k containers must follow the same path.

Problem 17 (2 point) For each commodity k in MEDITLINE, write up a table with all possible routes from s_k to t_k through the graph $G=(V,E,c,u)$. A route may only use edges where the capacity is at least d_k, and it needs to be simple (i.e. cycle-free). For each commodity k and each feasible route state the corresponding cost of sending the d_k units. (Hint: there should be 8 distinct routes.) ■

We solve the LP-relaxed problem through *column generation*. After a few iterations we have the *restricted master problem* on the form

$$
\begin{array}{lccccccccccl}
\min & c_1x_1 & + & c_2x_2 & + & c_3x_3 & + & c_4x_4 & & & & \\
\text{s.t.} & 3x_1 & & & & & + & 2x_4 & \leq & 4 & & (y_{AB}) \\
 & & & & & & & & \leq & 1 & & (y_{AD}) \\
 & & & & & & & & \leq & 5 & & (y_{BC}) \\
 & 3x_1 & & & & & + & 2x_4 & \leq & 4 & & (y_{BD}) \\
 & & & & & & & & \leq & 1 & & (y_{DB}) \\
 & & & & & 1x_3 & + & 2x_4 & \leq & 3 & & (y_{DC}) \\
 & & & & & & & & \leq & 3 & & (y_{DE}) \\
 & 3x_1 & & & & & + & 2x_4 & \leq & 5 & & (y_{EA}) \\
 & & & 3x_2 & & & & & \leq & 3 & & (y_{ED}) \\
 & x_1 & + & x_2 & & & & & \geq & 1 & & (\bar{y}_1) \\
 & & & & & x_3 & & & \geq & 1 & & (\bar{y}_2) \\
 & & & & & & & x_4 & \geq & 1 & & (\bar{y}_3) \\
 & x_1,x_2,x_3,x_4 \geq 0 & & & & & & & & & &
\end{array}
$$

where c_1, c_2, c_3, c_4 are some costs we found in previous problem. Let y_{ij} denote the dual variable corresponding to the capacity constraint for edge (i, j), and let $\overline{y}_k$ denote the dual variable corresponding to the demand constraint for commodity k.

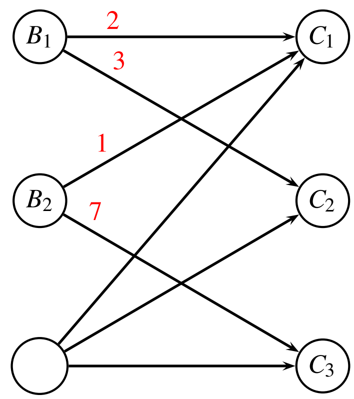
The dual variables for the above model are: $y_{AB} = -2$, while all other are $y_{ij} = 0$. Furthermore, $\overline{y}_1 = 21$, $\overline{y}_2 = 8$, $\overline{y}_3 = 30$.

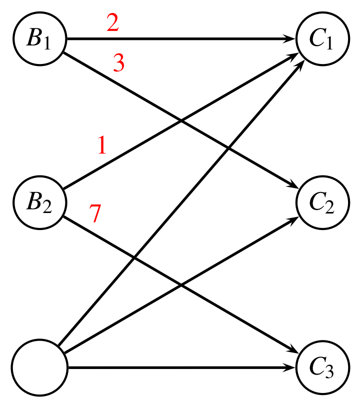
Problem 18 (4 point) For a given commodity k describe how a route with most negative reduced cost can be found using a shortest path algorithm. Draw the network for each commodity k (nodes, edges, costs, start-node, end-node), and report the optimal solution to the shortest-path problem, as well as the reduced cost. ■

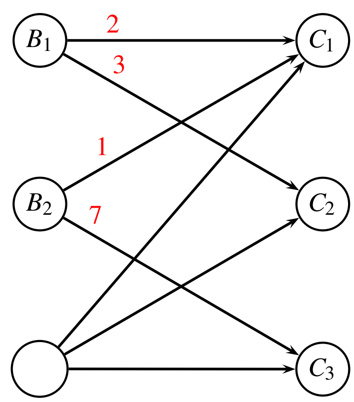
Problem 19 (1 point) Which path should be added to the restricted master problem? ■

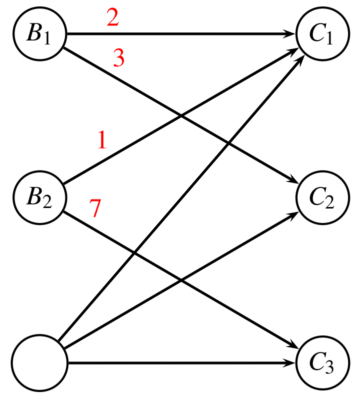
end of assignment

Transportation problem **Study number:**





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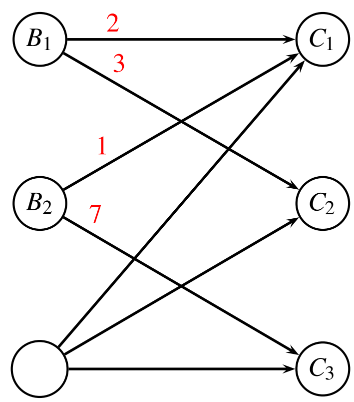
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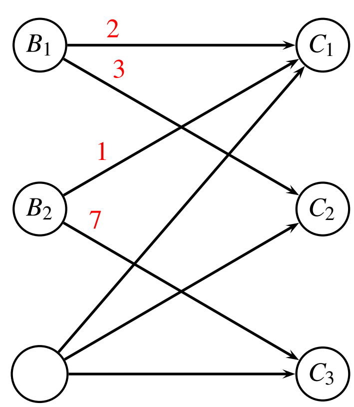
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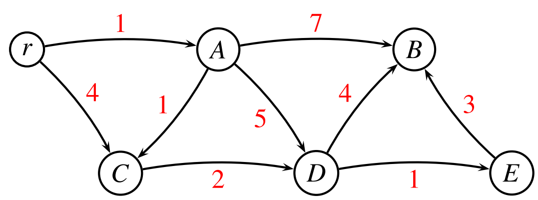
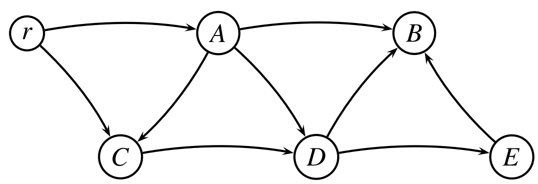
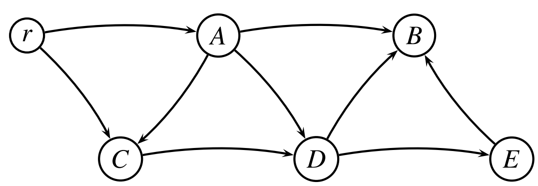
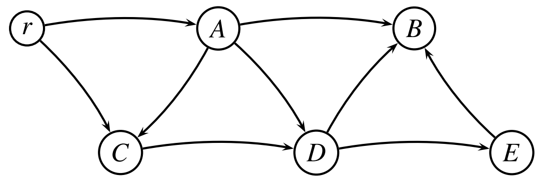
Shortest path problem **Study number:**





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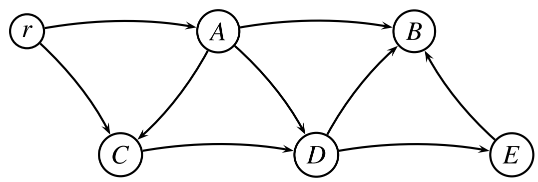
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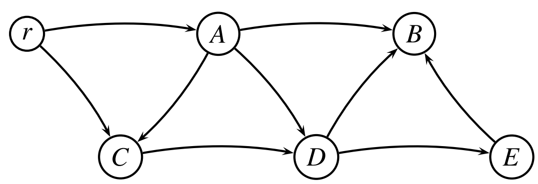
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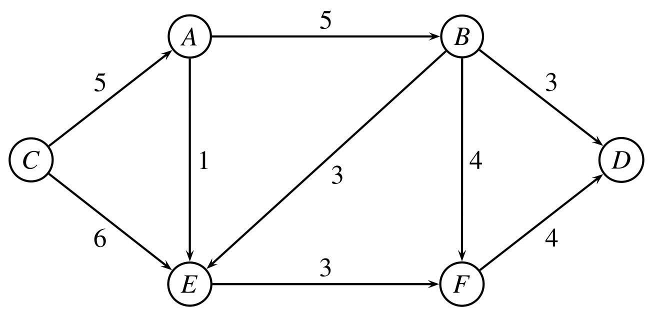
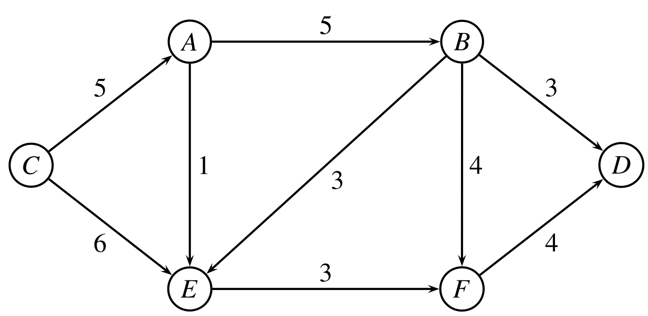
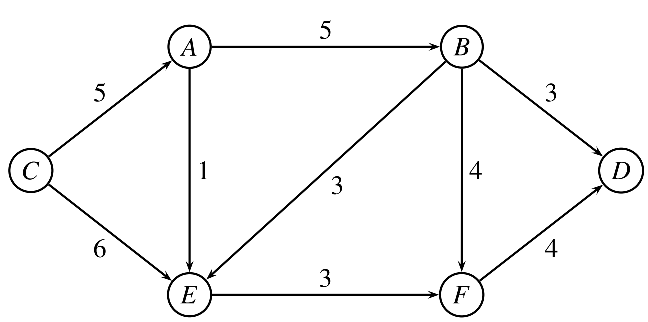
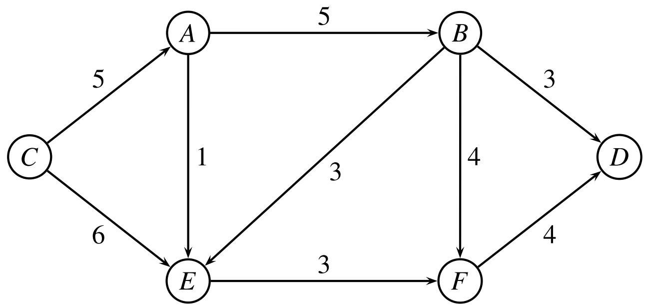
Circulation problem **Study number:**





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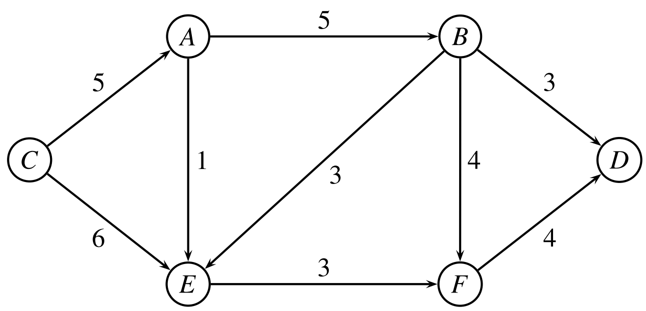
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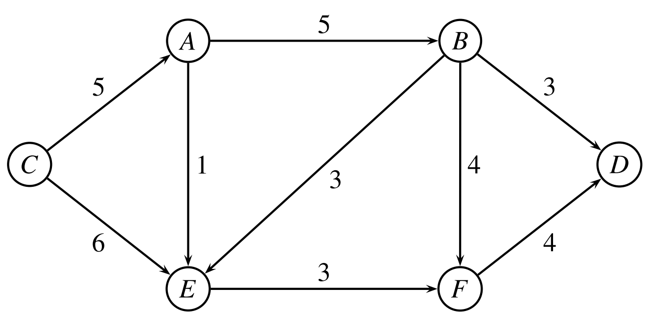
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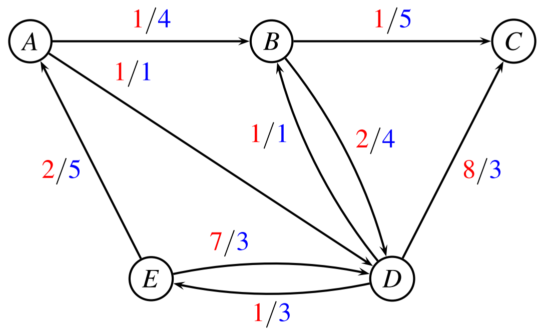
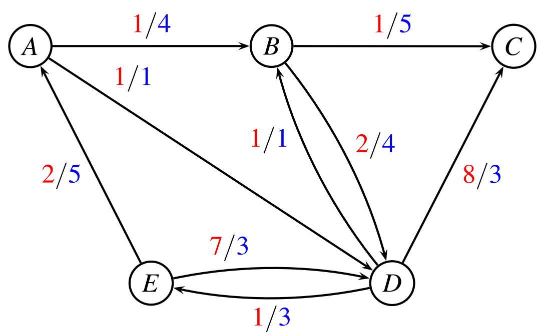
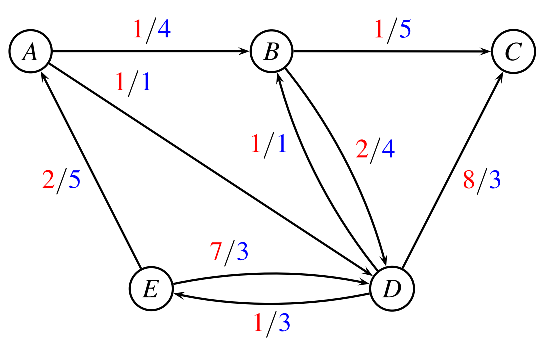
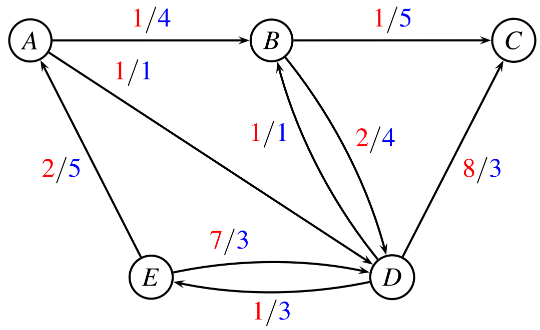
Repositioning of empty containers **Study number:**





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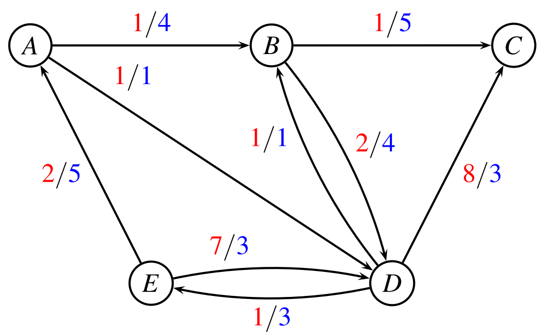
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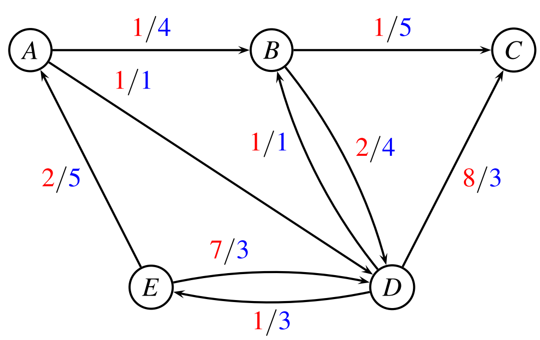
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Flowing containers **Study number:**